

# AARUSH SINGH

Graduate Software Engineer | Full-Stack · Cloud · IT Support | Melbourne, Australia  
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## SUMMARY

Software Engineering graduate from Swinburne University (First Class Honours) with experience building production systems, resolving technical incidents, and supporting end users across IT and operations environments. Currently providing IT field support across Swinburne teaching spaces while actively seeking a full-time graduate role. Skilled across the full stack from React and FastAPI to PostgreSQL and AWS, with a genuine interest in writing software that works reliably in the real world.

## EDUCATION

### Swinburne University of Technology

Feb 2022 – Dec 2025

*Bachelor of Engineering (Honours) – Software Engineering | GPA: 3.2 / 4.0 | Melbourne, Australia*

**Coursework:** Distributed Systems, Cloud Computing (AWS), Database Management, Computer Networks, Software Architecture, Linux Systems

**Awards:** International Excellence Scholarship (75%), Industry Experience Certificate, Swinburne Emerging Leader, Sustainability Certificate

## TECHNICAL SKILLS

**Languages:** Python, JavaScript (ES6+), TypeScript, SQL, Java, C#, C++, HTML5, CSS3

**Web & Frameworks:** React.js, React Native, FastAPI, Node.js/Express, Flask, ROS2, Tailwind CSS

**Databases:** PostgreSQL (Supabase), MySQL, MongoDB, Firebase – schema design, querying, data modelling

**Cloud & DevOps:** AWS (CloudWatch, EC2, S3), Microsoft Azure, Docker, GitHub Actions (CI/CD), Vercel, Render, Git

**IT Support & Tools:** Hardware troubleshooting, AV systems, Jira, ServiceNow, incident triage, escalation documentation

**AI / ML:** CNNs, LSTMs, sensor fusion, OCR (Tesseract, EasyOCR), TensorFlow, OpenCV, Claude API, GitHub Copilot

**APIs & Integration:** RESTful APIs, Webhooks, JWT, Stripe, Postman, HTTP/HTTPS debugging, TLS/SSL, PKI

**Testing & Quality:** pytest, Jest, integration testing, unit testing, code review

## WORK EXPERIENCE

### IT Delivery Consultant

Jan 2026 – Present

*Swinburne University of Technology, Infrastructure Services & Operations | Hawthorn, VIC*

- Conduct daily readiness checks across teaching spaces and computer labs, verifying power-on status, network connectivity, and AV functionality for all workstations before classes begin; log all identified faults in the ticketing system with clear reproduction steps and escalation notes to ensure issues are resolved within service timelines.
- Provide first-line IT support to academic staff and students across in-person, phone, and email channels, resolving hardware, software, and AV incidents on-site and escalating complex cases to senior IT staff with thorough documentation.
- Contribute to team knowledge by documenting recurring incidents and their resolutions in internal knowledge base articles, reducing repeat ticket volume and improving handover quality across shifts.

### Final Year Project Engineer – Autonomous Robotics

Feb 2025 – Nov 2025

*Swinburne University of Technology, Faculty of Engineering | Melbourne, VIC*

- Developed autonomous navigation algorithms in Python and C++ using ROS2 Nav2, achieving 95%+ accuracy on real robot hardware through iterative real-world testing across a warehouse simulation environment.
- Designed a sensor fusion pipeline integrating LiDAR, wheel encoders, and camera data with real-time filtering and anomaly detection, maintaining stable localisation under dynamic conditions; built a dual-engine OCR system (Tesseract and EasyOCR) with confidence scoring and fuzzy matching for reliable warehouse signage recognition.
- Diagnosed cross-subsystem failures spanning SLAM, navigation, and sensor layers; delivered structured technical progress reports and corrective action summaries to a senior academic client across a 10-month engagement.

### Software Engineer Intern

Jul 2024 – Feb 2025

*YEMECCompany Pty Ltd | AI-powered fintech platform – split payments and loyalty automation | Melbourne, VIC*

- Diagnosed and resolved live production incidents involving Stripe API failures and webhook synchronisation errors by analysing AWS CloudWatch logs and tracing HTTP request flows; tracked all defects in Jira with reproduction steps, severity assessments, and escalation notes for senior engineers.
- Built and maintained RESTful API endpoints and webhook integrations in Node.js and Express with a PostgreSQL backend; delivered a vendor dashboard with real-time order updates and Stripe payment processing, and configured Docker, GitHub Actions CI/CD, and CloudWatch monitoring across the production environment.
- Shipped a full React Native mobile application end-to-end covering vendor authentication, flyer customisation, and loyalty reward flows; served as the primary technical contact for vendor stakeholders, translating incident findings into plain language and maintaining progress throughout resolution.

## Airport Operations Specialist

Jun 2022 – Present

Uber | Melbourne Airport, VIC

- Delivered on-ground and in-app technical support to 100+ riders and drivers daily across Melbourne Airport terminals, resolving app issues including payment failures and authentication errors; compiled structured rider feedback reports that contributed to operational improvements rolled out across ANZ and APAC.
- Monitored Uber's Perfect Match bay-allocation system during peak shifts to ensure correct rider-to-bay assignments, reducing friction at high-volume pick-up points; supported the ground-level rollout of Uber Kiosks at MEL, Uber Child Seat, and UberMax – all piloted from Melbourne before expanding across ANZ.
- Managed real-time queue flow across multiple terminals during peak periods and provided structured feedback to the operations team on newly introduced technology and process changes, directly informing product and experience improvements deployed at network scale.

## PROJECTS

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**Full-Stack Restaurant Ordering Platform** | React, FastAPI, PostgreSQL, Stripe, Cloudinary, Vercel

[mohrstreetcafe.com.au](https://mohrstreetcafe.com.au)

- Designed and shipped a complete production ordering system for a live cafe, featuring QR table ordering, real-time order tracking, an admin dashboard, 30+ delivery zones, customer loyalty points, thermal receipt printing, and Stripe payment integration.
- Architected a PostgreSQL schema on Supabase to manage orders, menus, delivery zones, and loyalty data; deployed across Vercel and Render with Cloudinary for media storage, maintaining zero-downtime releases throughout.
- Handled all ongoing production responsibilities independently – incident monitoring, bug triage, schema updates, and feature releases – for a system actively serving real customers.

**Full-Stack E-Commerce Web Application** | Flask, React.js, MySQL, Tailwind CSS

[GitHub](#)

- Built a retail operations platform covering product catalogue browsing, inventory management, order tracking, and separate authentication flows for staff and customers.
- Designed a normalised MySQL schema managing 20+ product records with consistent data integrity across stock levels, order history, and user accounts.
- Connected a Flask REST backend to a React and Tailwind CSS frontend, then validated the full-stack architecture with integration tests across all core user flows before deployment.

**Facial Recognition Attendance System** | Python, Flask, React, OpenCV, CNN, TensorFlow

[GitHub](#)

- Built a secure attendance management system using CNN-based facial recognition with anti-spoofing through blink detection and emotion recognition via OpenCV, achieving 95%+ liveness verification accuracy on test data.
- Developed a Flask REST backend to manage user authentication, registration, and attendance records, with a React frontend handling live webcam capture and user enrolment.
- Validated the full-stack architecture with integration tests across enrolment, recognition, and attendance flows before deployment to ensure reliability in a real-world environment.

**Full-Stack Personal Finance Web App** | React.js, Node.js, MongoDB, Azure, Docker, GitHub Actions

[GitHub](#)

- Built a personal finance application with real-time analytics dashboards, interactive spend visualisations by category, and per-user data isolation via JWT-based authentication with protected routes.
- Containerised the application with Docker and deployed to Microsoft Azure via an automated GitHub Actions CI/CD pipeline, achieving zero-downtime releases across updates.
- Configured production monitoring with logging, health checks, and alerting to surface issues early, with a MongoDB backend handling all CRUD operations through RESTful API endpoints.

## CERTIFICATIONS & WORK RIGHTS

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**Temporary Graduate Visa (Subclass 485)** – Full work rights, valid until 19 March 2028

**Working With Children Check (WWCC)** – Current and valid

**Python 3** – Udemy, 2020

**Python** – Kaggle, 2020

**Intro to SQL** – Kaggle, 2020

**Intro to Machine Learning** – Kaggle, 2020

*References available upon request.*